

Internet Resources

Kinetochore Function. McGill University. <<http://www.sus.mcgill.ca/bio202/00b/lec2/sld028.htm>>.

Kinetochore Structure. University of Texas. <<http://www.esb.utexas.edu/dr325/Supplements/kinetchr.htm>>.

polypeptide chain of amino acids

Chaperones

Molecular chaperones are proteins and protein complexes that bind to misfolded or unfolded **polypeptide** chains and affect the subsequent folding processes of these chains. All proteins are created at the ribosome as straight chains of amino acids, but must be folded into a precise, three-dimensional shape (conformation) in order to perform their specific functions. The misfolded or unfolded polypeptide chains to which chaperones bind are said to be “non-native,” meaning that they are not folded into their functional conformation. Chaperones are found in all types of cells and cellular compartments, and have a wide range of binding specificities and functional roles.

Discovery of Chaperones

Chaperones were originally identified in the mid-1980s from studies of protein folding and assembly in plant chloroplasts. A new protein was identified that was required for correct folding of a large enzyme complex in chloroplasts, yet the mysterious protein was not associated with the final assembled complex. It was quickly determined that this “chaperone” protein directing correct assembly was identical to one of the many proteins expressed at high levels when cells are grown at high temperatures (hence the common alternative name, “heat-shock protein,” or Hsp).

It was later discovered that chaperones recognize the non-native, partially misfolded states of proteins that accumulate during high temperature stress. Most chaperones are also abundantly expressed under normal cell growth conditions, where they recognize non-native conformations occurring during both protein synthesis (prior to correct polypeptide chain folding), and later misfolding events.

Recognizing and Correcting Mistakes

Careful study, both **in vivo** and in the test tube, has demonstrated that molecular chaperones bind to their non-native substrate proteins by recognizing exposed **non-polar** surfaces (“non-polar” means that they are not attracted to water). In correctly folded proteins, these surfaces are usually buried away from the watery environment surrounding the protein. Chaperones promote correct folding of their substrate proteins by unfolding incorrect polypeptide chain conformations, and, in some cases, by providing a sequestered environment in which correct protein folding can occur. The activity of chaperones often requires the binding and hydrolysis of adenosine triphosphate (ATP).

Although only 20 to 30 percent of polypeptide chains require the assistance of a chaperone for correct folding under normal growth conditions, molecular chaperones are absolutely required for cell viability. Discussed

in vivo “in life”; in a living organism, rather than in a laboratory apparatus

non-polar without charge separation; not soluble in water

below are a few of the most common classes of molecular chaperones and their effects on protein folding in the cell.

Two Common Chaperone Systems: Hsp70 and Hsp60

Hsp70 chaperones (so called because their size is approximately 70,000 daltons, or atomic mass units) are a very large family of proteins whose amino acid sequences are very similar, indicating how important their structure is to their function. A single cell or cellular compartment may contain multiple Hsp70 chaperones, each with a specific function. In addition, the Hsp70 chaperones often work in concert with one or more smaller co-chaperone proteins, which serve to modulate the activity of the chaperone.

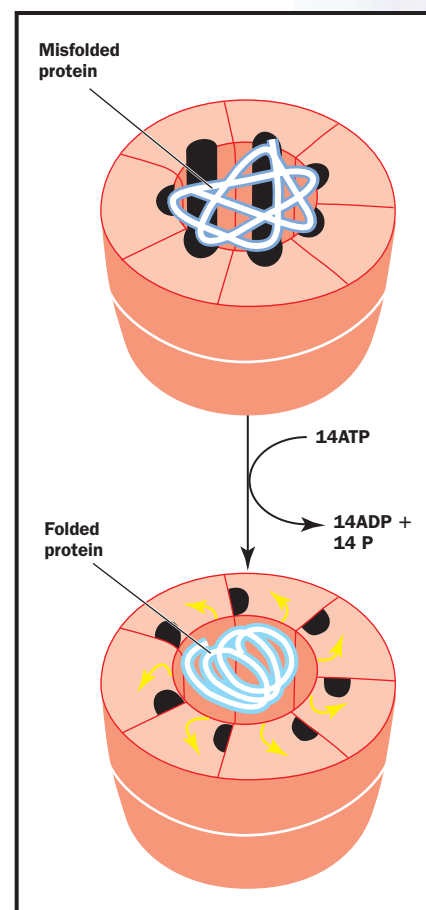
Some of the well-studied Hsp70 chaperones include DnaK from the bacterium *Escherichia coli*, the Ssa and Ssb proteins from yeast, and BiP (for “binding protein”) from the mammalian **endoplasmic reticulum**. Hsp70 chaperones are often located where unfolded polypeptide chains typically appear. For example, Ssb chaperones associate with ribosomes, so that they are close to newly synthesized, unstructured polypeptide chains. It is thought that the binding of Hsp70 chaperones to these unfolded chains prevents inappropriate partial folding until the entire polypeptide chain is available for correct folding.

Hsp60 chaperones (also called “chaperonins”) are barrel-shaped structures composed of fourteen to sixteen subunits of proteins that are approximately 60,000 daltons in size. Each subunit has a patch of non-polar amino acid groups lining the inner surface of the barrel; this patch recognizes the exposed non-polar amino acids of misfolded proteins. The binding and **hydrolysis** of ATP triggers conformational changes within the barrel, which (1) unfolds the misfolded conformation and releases the unfolded chain into the center of the barrel, (2) closes the top of the barrel with the binding of a co-chaperone “cap,” and thereby (3) provides a protected environment in which correct folding can occur. Upon dissociation of the co-chaperone, the fully or partially folded protein is released into the general cellular environment.

The most extensively studied Hsp60 chaperones include GroEL from *E. coli* and TRiC/CCT from eukaryotic cells. GroEL appears to function as a general chaperone and interacts with 10 to 15 percent of all *E. coli* polypeptide chains, with a definite bias toward proteins that are small enough to fit within its central cavity. TRiC/CCT recognizes a much smaller set of proteins, and appears to play an additional role in the assembly of multiprotein complexes.

Other Chaperone Systems

While Hsp70 and Hsp60 chaperones are the most extensively studied chaperone systems, there are many other chaperones with distinct cellular functions. These functions include modifying polypeptides after formation by altering the bonds within and between chains. It appears that some chaperones, in addition to attempting to rescue partially misfolded proteins, also alert the protein degradation system of the cell to the presence of substrate proteins that are too misfolded for rescue. It is expected, with the explosion of information provided by genome sequencing efforts, that many additional chaperones will be identified in the near future.

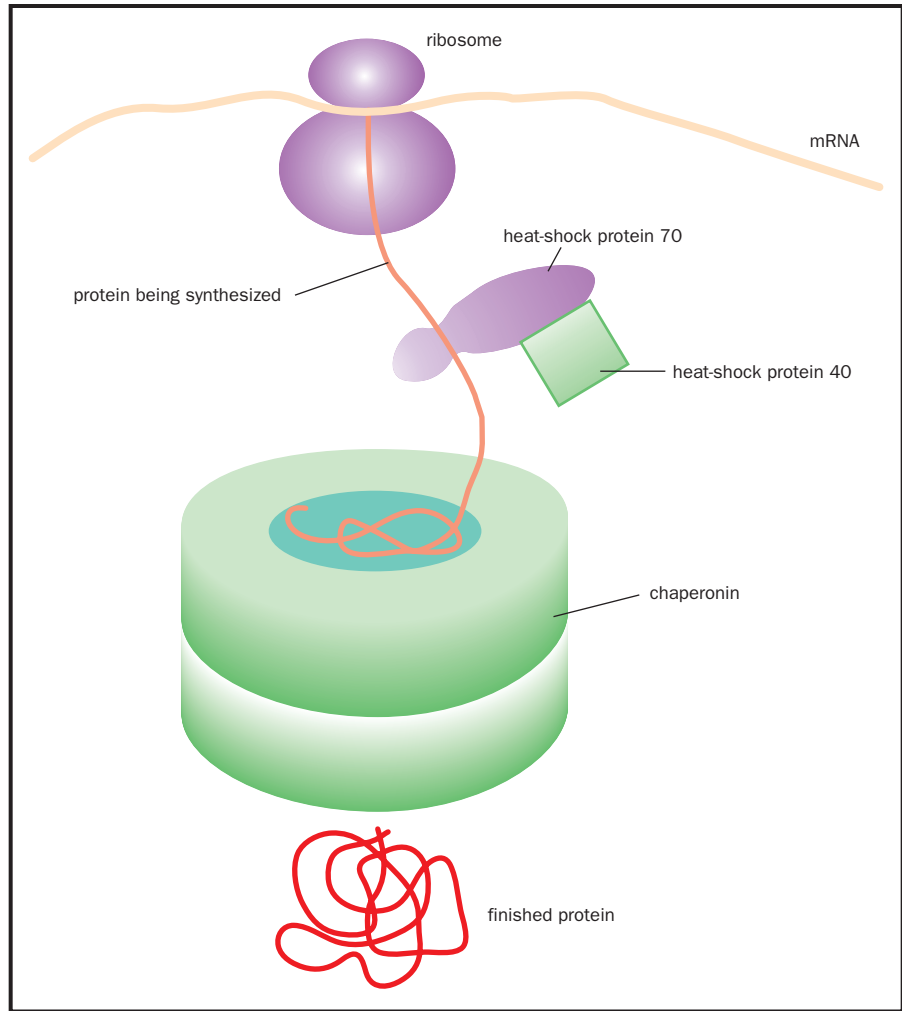


Chaperones use energy from ATP to help misfolded proteins refold properly.

endoplasmic reticulum
network of membranes within the cell

hydrolysis splitting with water

Heat-shock proteins and chaperonins cooperate to fold newly synthesized proteins. The correct three-dimensional conformation is essential for protein function. Adapted from <http://www.nurseminerva.co.uk/chaperon.htm>.



Chaperones and Human Disease

It is clear that molecular chaperones assist with the folding of newly synthesized proteins and correct protein misfolding. Recent studies now suggest that defects in molecular chaperone/substrate interactions may also play a substantial role in human disease. For example, mutations linked to Alzheimer's disease have been shown to disrupt the expression of chaperones in the endoplasmic reticulum. In addition, several genes linked to eye degeneration diseases have recently been identified as putative molecular chaperones. SEE ALSO CELL, EUKARYOTIC; POST-TRANSLATIONAL CONTROL; PROTEINS; RIBOSOME; TRANSLATION.

Patricia L. Clark

Bibliography

- Frydman, Judith. "Folding of Newly Translated Proteins In Vivo: The Role of Molecular Chaperones." *Annual Review of Biochemistry* 70 (2001): 603–647.
- Wickner, Sue, Michael R. Maurizi, and Susan Gottesman. "Posttranslational Quality Control: Folding, Refolding, and Degrading Proteins." *Science* 286 (1999): 1888–1893.

Internet Resources

"Chaperone." Nurse Minerva. <<http://www.nurseminerva.co.uk/chaperon.htm>>.

"Innovations." Environmental Health Perspectives. National Institutes of Health. <<http://ehpnet1.niehs.nih.gov/docs/1994/102-6-7/innovations.html>>.

"Molecular Chaperones." Federation of American Societies for Experimental Biology. <<http://www.faseb.org/opar/protfold/molechap.html>>.

Chromosomal Aberrations

Chromosomal aberrations are abnormalities in the structure or number of chromosomes and are often responsible for genetic disorders. For more than a century, scientists have been fascinated by the study of human chromosomes. It was not until 1956, however, that it was determined that the actual diploid number of chromosomes in a human cell was forty-six (22 pairs of **autosomes** and two sex chromosomes make up the human genome). In 1959 two discoveries opened a new era of genetics. Jerome Lejeune, Marthe Gautier, and M. Raymond Turpin discovered the presence of an extra chromosome in Down syndrome patients. And C. E. Ford and his colleagues, P. A. Jacobs and J. A. Strong first observed sex chromosome anomalies in patients with sexual development disorders.

autosomes chromosomes that are not sex determining (not X or Y)

Advances in Chromosomal Analysis

Identification of individual chromosomes remained difficult until advances in staining techniques such as Q-banding revealed the structural organization of chromosomes. The patterns of bands were found to be specific for individual chromosomes and hence allowed scientists to distinguish the different chromosomes. Also, such banding patterns made it possible to recognize that structural abnormalities or aberrations were associated with specific genetic syndromes. Chromosome disorders, or abnormalities of even a minute segment (or band) are now known to be the basis for a large number of genetic diseases.

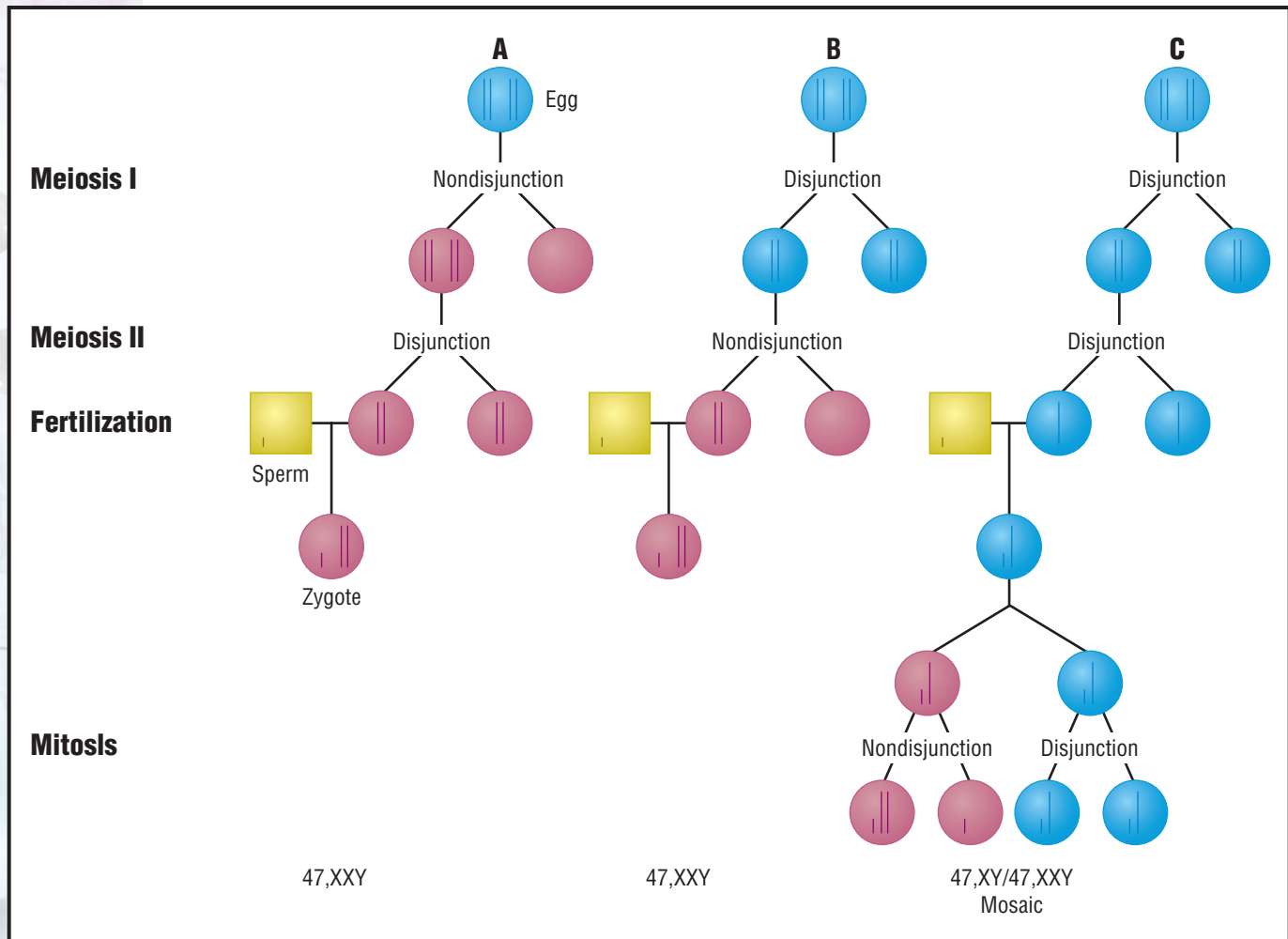
Chromosomal disorders and their relationship to health and disease are studied using the methods of **cytogenetics**. Cytogenetic analysis is now an integral diagnostic procedure in prenatal diagnosis. It is also utilized in the evaluation of patients with mental retardation, multiple birth defects, and abnormal sexual development, and in some cases of infertility or multiple miscarriages. Cytogenetic analysis is also useful in the study and treatment of cancer patients and individuals with hematologic disorders. The types of chromosomal abnormalities that can be detected by cytogenetics are numerical aberrations, translocations, duplications, deletions, and inversions.

cytogenetics study of chromosome structure and behavior

Chromosomal Aberrations

Chromosomal abnormalities can result from either a variation in the chromosome number or from structural changes. These events may occur spontaneously or can be induced by environmental agents such as chemicals, radiation, and ultraviolet light. However, mutations are most likely due to mistakes that occur when the genes are copied as the cells are dividing to produce new cells. These abnormalities may involve the autosomes, sex chromosomes, or both. The disruption of the DNA sequence or an excess or deficiency of the genes carried on the affected chromosomes results in





gametes reproductive cells, such as sperm or eggs

a mutation. Such a change may or may not alter the protein coded by a gene. Often, however, a mutation results in the disruption of gene functionality. The resulting altered or missing protein can disrupt the way a gene is meant to function and can lead to clinical disease. Only mutations occurring to the DNA in the **gametes** will potentially pass on to the offspring.

Mutations appear in gametes in one of two ways. A mutation may be inherited from one of an individual's parents. However, a mutation may also occur for the first time in a single gamete, or during the process of fertilization between an egg cell and a sperm cell. In this case the mutation or change is often called a *de novo* mutation. The parents are not affected by the condition and are not "carriers" of the mutation. The affected individual will have this mutation in all of his or her cells and may be able to pass the mutation on to any offspring. Some common abnormalities and their resulting phenotypes are discussed below.

Aneuploidy

Aneuploidy is the gain or loss of individual chromosomes from the normal diploid set of forty-six chromosomes. As in structural anomalies, the error may be present in all cells of a person or in a percentage of cells. Changes in chromosome number generally have an even greater effect upon survival

than changes in chromosome structure. Considered the most common type of clinically significant chromosome abnormality, it is always associated with physical and/or mental developmental problems. Most aneuploid patients have a **trisomy** of a particular chromosome. **Monosomy**, or the loss of a chromosome, is rarely seen in live births. The vast majority of monosomic embryos and fetuses are probably lost to spontaneous abortion during the very early stages of pregnancy. An exception is the loss of an X chromosome, which produces Turner's syndrome. Trisomy may exist for any chromosome, but is rarely compatible with life.

Aneuploidy is believed to arise from a process called nondisjunction. Nondisjunction occurs when chromosomes do not separate correctly during meiosis. The direct result is that one gamete will have an extra chromosome and the other will be lacking a chromosome. When these gametes are fertilized by a normal gamete, they have either an extra chromosome (trisomy) or are missing a chromosome (monosomy).

Disorders Associated with Aneuploidy

Three well-known autosomal chromosome disorders associated with trisomies of entire autosomes are sometimes found in live births. These are trisomy 21 (Down syndrome), trisomy 13, and trisomy 18. Growth retardation, mental retardation, and multiple congenital anomalies are associated with all three trisomies. However, each has distinctive morphological characteristics, which are presumably determined by the extra dosage of the specific genes on the additional chromosome.

Down syndrome (chromosome 21) is the most frequent trisomy found in humans, and one of the most common conditions encountered in genetic counseling. General characteristics are mental retardation, distinctive palm prints, and a common facial appearance. The average life expectancy is now much greater thanks to improvements in medical care. Generally, individuals with Down syndrome have affable personalities and are able to be partially independent. The incidence of Down syndrome is about 1 in 800 children and is often associated with later maternal age (as may also be the case with other aneuploids).

Down syndrome appears to be related to the difference in gamete formation (gametogenesis) between males and females. In females, **oocytes** are formed before birth and held in a static state until ovulation. In the case of older mothers, an oocyte may be in this stage for more than forty years, during which time environmental factors may affect the genetic material. In trisomy 13 and trisomy 18 patients, congenital abnormalities are much more severe. These individuals generally do not live much beyond birth. Both trisomy 13 and trisomy 18 result in syndromes characterized by specific dysmorphic features and severe organ malformations.

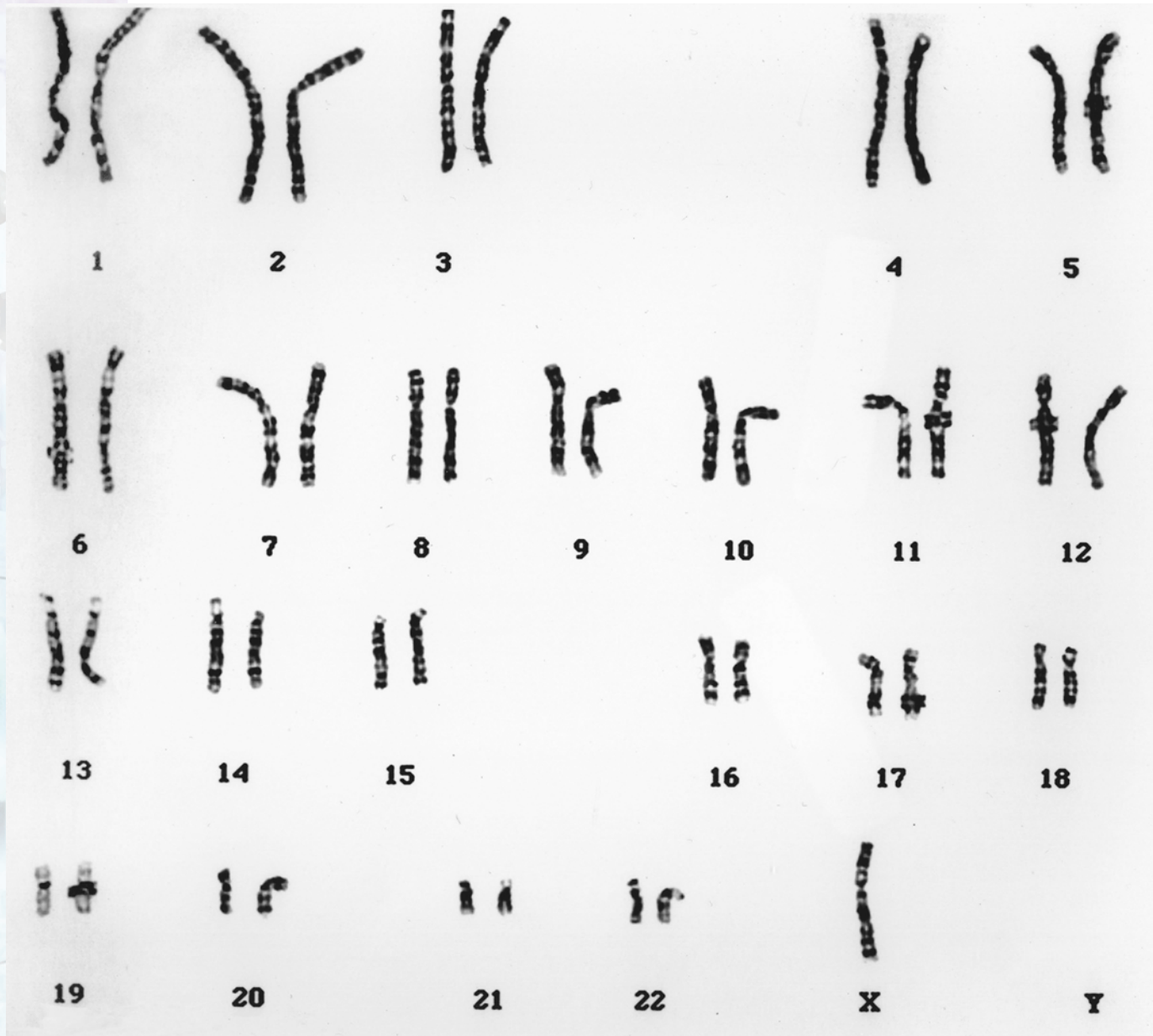
In addition to trisomies involving the autosomal chromosomes, aneuploidy may also involve the sex chromosomes. Two examples are Turner's syndrome and Klinefelter's syndrome. As mentioned previously, Turner's syndrome is a monosomy involving the X chromosomes. Turner's syndrome females possess forty-five chromosomes (45, X) as compared to clinically normal forty-six (46, XX). They are usually sterile and short in stature with some neck webbing. Klinefelter's syndrome patients have a trisomy involving the

trisomy presence of three, instead of two, copies of a particular chromosome

monosomy gamete that is missing a chromosome

oocytes egg cells





This karyotype of Turner's syndrome is characterized by a missing X chromosome.

sex chromosomes and thus have forty-seven chromosomes (47, XXY). Klinefelter's syndrome individuals are sterile males possessing some female characteristics. These chromosome abnormalities are of interest especially for their implications in infertility and abnormal development.

Abnormalities of Chromosomal Structure

Four types of structural changes may occur in chromosomes: duplications, deletions, translocations, and inversions. All may result when there is breakage of the chromosomes and a rejoining or loss of chromosome fragments. If the same broken ends rejoin, the chromosome becomes intact once again. The resulting effects of such events depend on how large they are and where they occur on the chromosome. Rearrangements may occur in many forms and are less common than abnormalities of chromosome number.

The most common type of rearrangement is called a balanced **translocation** because the amount of genetic information within that cell is normal even though it is repositioned. Therefore the individual with a balanced translocation may appear normal. However, there will be a risk to the children of a carrier of a balanced translocation since that person is likely to produce unbalanced gametes (bearing too little or too much genetic information), and therefore the risk of having abnormal offspring is increased. Rearrangements such as aneuploidy may be found in all cells of an individual, or they may occur only in a percentage of an individual's cells. This latter condition is known as mosaicism. In general, mosaic individuals show a less severe expression of their syndrome than those with chromosome abnormalities in all their cells.

translocation movement of a chromosome segment from one chromosome to another

Unbalanced Chromosome Rearrangements

A rearrangement is considered unbalanced if it results in extra or missing information. Structural rearrangements may be caused by a number of factors including chemicals, some viral infections, and ionizing radiation. Because the complement of DNA or genetic material in the chromosomes is greater or less than the complement of DNA in a normal set of chromosomes, there is likely to be abnormal development.

Deletions

A deletion is the loss of a segment of a chromosome. The amount of deleted material may be any length from a single base to a large piece of the chromosome. The result is a chromosomal imbalance, with the individual being monosomic or possessing half of the required genes present in a normal individual for the segment of DNA missing. Only small deletions are tolerated, and the effect on the individual will depend upon the size of the deleted segment and the number and functionality of the genes that are contained within it. Larger deletions and the deletion of an entire chromosome always result in nonviable embryos. *Cri du Chat's* ("cat's cry") syndrome individuals have a deletion of the short arm of chromosome 5. Although they possess the usual signs of chromosomal anomalies, such as mental retardation and low birth weight, their appearance is not extraordinarily different from normal individuals. One peculiarity is that affected infants make an unusual cry resembling that of a cat, hence the name of the syndrome. Two other interesting diseases are Prader-Willi's syndrome and Angelman's syndrome. In both cases, patients with these diseases possess a deletion in the long arm of chromosome 15. Interestingly, the deletion is in the same location, but the resulting syndrome depends on whether the deletion was in the maternal or paternal chromosome.

Duplications

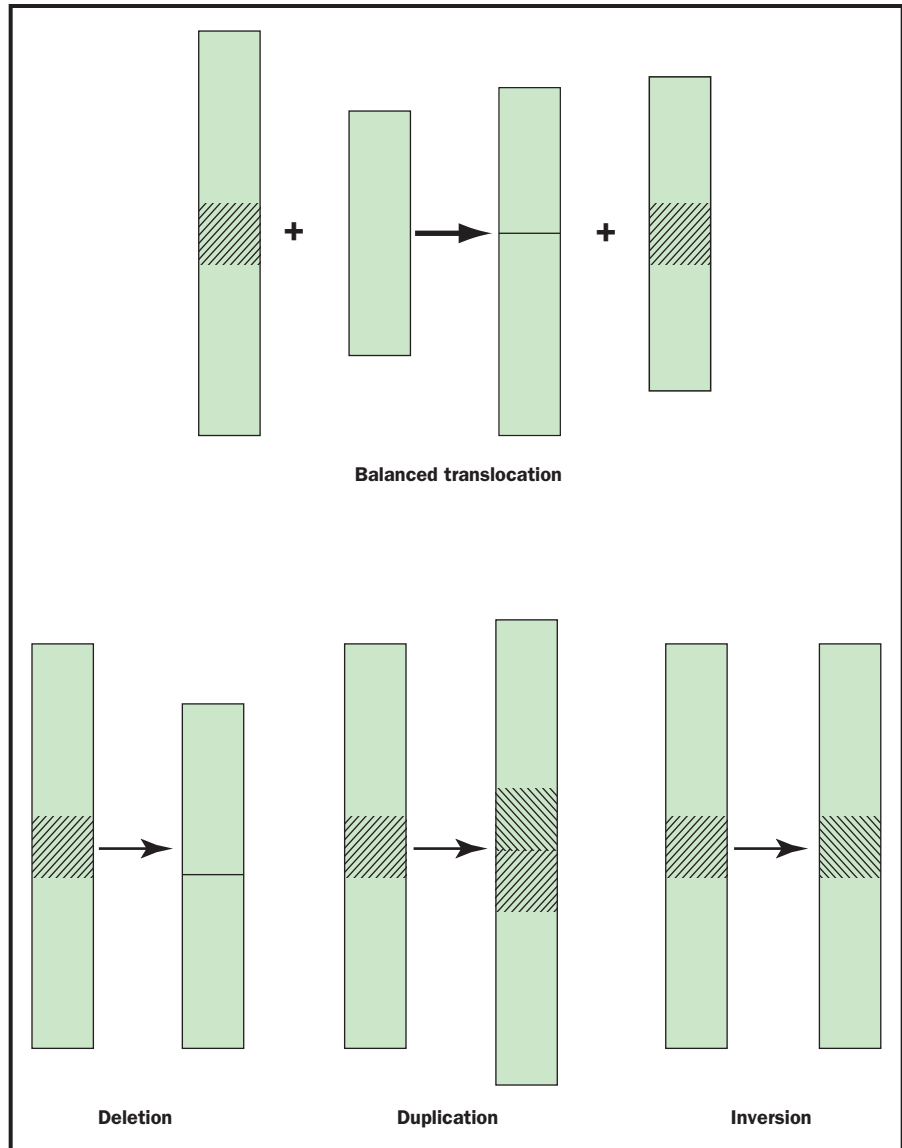
Duplications also result from the reuniting of broken pieces of **homologous** chromosomes. In some cases the chromosome pieces rejoin in such a way that there is a doubling, or redundancy, of a portion of the chromosome. This changes the number of genes present and may result in a problem with health, development, or growth.

homologous similar in structure

Large insertions and deletions prevent the production of useful proteins. The effect of smaller insertions or deletions depends upon how many bases



Schematic representation of four types of chromosomal aberrations.



are involved. Sometimes an entire gene can be inserted (in duplications) or deleted. The effect depends upon where in the genome the changes occur and how many base pairs are involved.

Inversions

An inversion is the rotation of a broken chromosome segment in such a way that it rejoins the chromosome in a reversed state, or is flipped, end to end. Inversions are usually characterized by whether the centromere is included in the inverted segment. Inversions containing the centromere are called pericentric. Those not containing the centromere are called paracentric. Although an inversion does not change the overall content of cellular DNA and can be considered a balanced translocation, it can affect a gene at many levels because it alters the normal DNA sequence. The gene may not produce its corresponding protein at all, or a nonfunctioning protein may result. There is a common inversion seen in human chromosomes involving chromosome 9. A small pericentric inversion is present in approximately 1

percent of tested individuals. There appears to be no detrimental effect on the carrier, and it does not appear to cause miscarriage or unbalanced offspring.

Recurrence Risk

Chromosomal aberrations may be inherited from a parent, and because of this many families seek genetic counseling in order to determine if a genetic disorder will recur in another member of the same generation or in generations that will follow. The family needs to know the genetic risk, also known as the recurrence risk, and any means by which transmission may be prevented. A recurrence risk will be calculated based on the accuracy of the diagnosis, the pedigree of the family, and the known genetic mechanisms of the disorder in question. SEE ALSO BIRTH DEFECTS; CHROMOSOME BANDING; CROSSING OVER; DOWN SYNDROME; MEIOSIS; MUTATION; NONDISJUNCTION; PRENATAL DIAGNOSIS.

Jacqueline Bebout Rimmer

Bibliography

- Cohen, Jon. "Sorting Out Chromosome Errors." *Science* (Jun. 21, 2002): 2164–2166.
- Haines, Jonathan L., and Margaret A. Pericak-Vance. *Approaches to Gene Mapping in Complex Human Diseases*. New York: John Wiley & Sons, 1998.
- Klug, William S., and Michael R. Cummings. *Concepts of Genetics*. Upper Saddle River, NJ: Prentice Hall, 1997.
- Thompson, Margaret W., Roderick R. McInnes, and Willard F. Huntington. *Genetics in Medicine*. Philadelphia: W. B. Saunders Company, 1991.
- Verma, Ram S., and Arvind Babu. *Human Chromosomes: Manual of Basic Techniques*. New York: Pergamon Press, 1989.
- Vogel, Friedrich, and Arno G. Motulsky. *Human Genetics*. Berlin, Germany: Springer-Verlag, 1986.
- Weaver, Robert F., and Philip W. Hedrick. *Genetics*. Dubuque, IA: W. M. C. Brown Publishers, 1989.
- Young, Ian D. *Introduction to Risk Calculation in Genetic Counseling*. Oxford, U.K.: Oxford University Press, 1999.

Chromosomal Banding

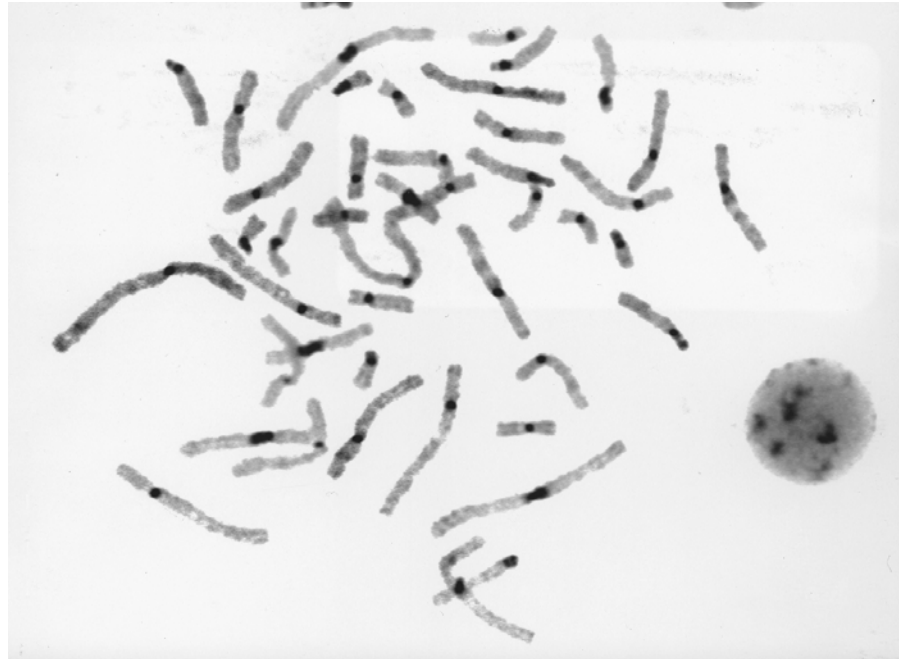
A chromosome banding pattern is comprised of alternating light and dark stripes, or bands, that appear along its length after being stained with a dye. A unique banding pattern is used to identify each chromosome and to diagnose chromosomal aberrations, including chromosome breakage, loss, duplication or inverted segments. In the 1950s, chromosomes from the cell's nucleus were identified with a uniform (unbanded) stain that allowed for the observation of the overall length and primary constriction (centromere) of each chromosome, as well as a secondary constriction in chromosomes 1, 9, 16 and the acrocentrics (chromosomes whose centromeres are near the tips). The staining techniques used to make the bands visible were developed in the late 1960s and early 1970s.

Chromosome Structure

To understand what chromosomal bands represent, it is helpful to understand the structure of chromosomes. Eukaryotic chromosomes are composed



C-banded metaphase cell.



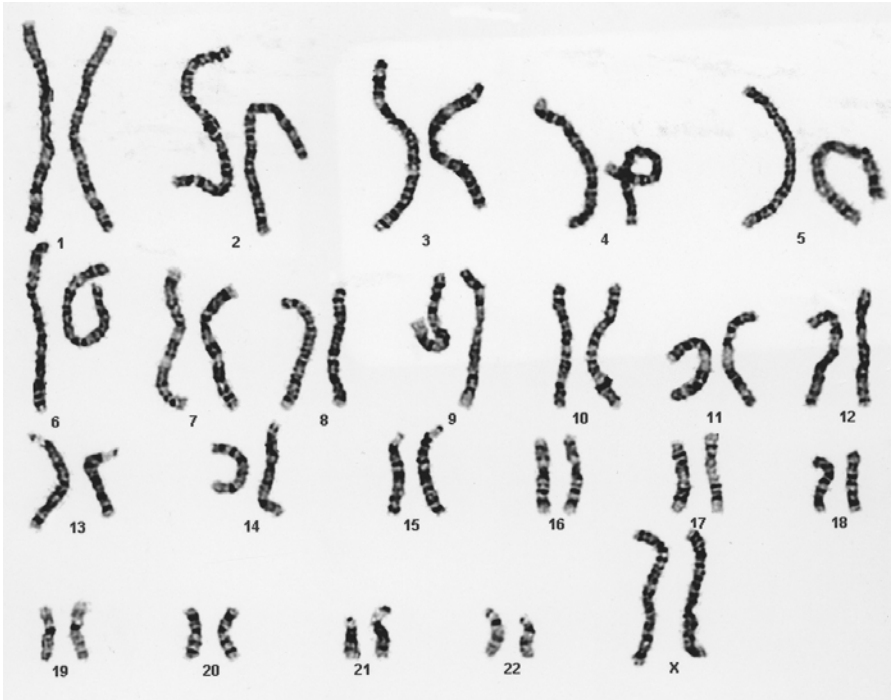
of chromatin, a combination of nuclear DNA and proteins. At metaphase, which is a phase in the cell cycle after the DNA in the nucleus has been replicated, each chromosome contains two identical strands of DNA. (Each strand contains two complementary strands of nucleotides.) The two strands of DNA, or chromatids, are arranged in a double-helix and are held together at a single point, the centromere, or primary constriction point.

mitosis separation of replicated chromosomes

metaphase stage in mitosis at which chromosomes are aligned along the cell equator

During **mitosis**, each chromatid becomes condensed approximately ten-thousand fold reaching maximal condensation at **metaphase**. DNA that was roughly 5 centimeters (2 inches) long is compacted to 5 micrometers. The DNA wraps around proteins called histones, forming complexes called nucleosomes. The nucleosomes twist around each other and assume a loop formation projecting out from the chromosome's protein backbone, or scaffold. The loops weave and condense further to package the DNA into a chromosome. Some of the looped segments of DNA remain close together and condense more than others, forming regions known as domains. These domains are the darkly-stained chromosomal bands that appear when specific stains are applied (such as Giemsa staining; see below).

Looped domains are also seen in polytene chromosomes, which are found mainly in insects of the order Diptera, including *Drosophila*, which are fruit flies. Polytene chromosomes are large chromosomes that are formed after DNA undergoes repeated rounds of replication without cell division. A polytene chromosome in a *Drosophila* salivary gland cell can contain as many as five thousand alternating dark and light bands. The dark bands correspond to the folded and looped DNA, and the lighter bands are composed of less condensed DNA. The DNA in polytene chromosomes becomes less condensed when genes become active, permitting DNA to be transcribed into messenger RNA. This unraveling is observed as "puffing" of the polytene chromosome. The puffing resolves (the DNA condenses again) as the genes become inactive.



G-banded karyotype from a normal female.

Chromosome Banding Techniques

Quinacrine mustard, an alkylating agent, was the first chemical to be used for chromosome banding. T. Caspersson and his colleagues, who developed the technique, noticed that bright and dull fluorescent bands appeared after chromosomes stained with quinacrine mustard were viewed under a fluorescence microscope. Quinacrine dihydrochloride was subsequently substituted for quinacrine mustard. The alternating bands of bright and dull fluorescence were called Q bands. Quinacrine-bright bands were composed primarily of DNA that was rich in the bases adenine and thymine, and quinacrine-dull bands were composed of DNA that was rich in the bases guanine and cytosine.

Other fluorescent dyes have been used to generate chromosomal banding patterns. The combination of the fluorescent dye, DAPI (4,6-Diamidino-2-Phenylindole) with a non-fluorescent counterstain, such as Distamycin A, will also stain DNA that is rich in adenine and thymine. It will particularly highlight regions that are on the Y chromosome, on chromosomes 9 and 16, and on the proximal short arms of the chromosome 15 **homologues**, or pair.

Giemsa has become the most commonly used stain in cytogenetic analysis. Staining a metaphase chromosome with a Giemsa stain is referred to as G-banding. Unlike Q-banding, most G-banding techniques require pre-treating the chromosomes with either salt or a proteolytic (protein-digesting) enzyme. “GTG banding” refers to the process in which G-banding is preceded by treating chromosomes with trypsin. G-banding preferentially stains the regions of DNA that are rich in adenine and thymine. In general, the bands produced correspond with Q-bright bands. The regions of the chromosome that are rich in guanine and cytosine have little affinity for the dye and remain light.

homologues chromosomes with corresponding genes that pair and exchange segments in meiosis



G-banded metaphase
from a normal female.



Standard G-band staining techniques allow between 400 and 600 bands to be seen on metaphase chromosomes. With high-resolution G-banding techniques, as many as two thousand different bands have been catalogued on the twenty-four human chromosomes. Jorge Yunis introduced a technique to synchronize cells so they are held at the same stage in the cell cycle. Cells are synchronized by making them deficient in folate, thereby inhibiting DNA synthesis. By rescuing the cells with thymidine, DNA synthesis is initiated and the timing of the prophase and prometaphase stages of the cell cycle can be predicted. Yunis's technique allows more bands to be resolved, as chromosomes produced from either prophase or prometaphase are less condensed and are thus longer than metaphase chromosomes.

Other Banding

R-banding is the reverse pattern of G bands so that G-positive bands are light with R-banding methods, and vice versa. R-banding involves pretreating cells with a hot salt solution that **denatures** DNA that is rich in adenine and thymine. The chromosomes are then stained with Giemsa. R-banding is helpful for analyzing the structure of chromosome ends, since these areas usually stain light with G-banding.

C-banding stains areas of heterochromatin, which is tightly packed and repetitive DNA. NOR-staining, where NOR is an abbreviation for "nucleolar organizing region," refers to a silver staining method that identifies genes for ribosomal RNA that were active in a previous cell cycle.

Fluorescence *In Situ* Hybridization

Fluorescence *in situ* hybridization (FISH) is a molecular cytogenetic technique that allows cytogeneticists to analyze chromosome resolution at the DNA or gene level. FISH can be performed on dividing (metaphase) and

denatures destroys the
structure of

non-dividing (interphase) cells to identify numerical and structural abnormalities resulting from genetic disorders.

In FISH, cytogeneticists utilize one or more FISH probes that typically fall into one of the following three categories:

1. Repetitive sequences, including alpha satellite DNA, that bind to the centromere of a chromosome;
2. DNA segments, representative of the entire chromosome, that will bind to and cover the entire length of a particular chromosome; and
3. DNA segments from specific genes or regions on a chromosome that have been previously mapped or identified.

A probe is “tagged” either directly, by incorporating fluorescent nucleotides, or indirectly, by incorporating nucleotides with attached small molecules, such as biotin, digoxigenin, or dinitrophenyl, to which fluorescent antibodies can later be bound. The probe and the chromosomes (from either the metaphase or interphase cells) that are being analyzed are denatured and allowed to bind or hybridize to one another. If necessary, **antibodies** with a fluorescent tag are applied to the cells. The cells are then viewed with a fluorescence microscope. The fluorescent signals represent the probe(s) that is bound to the chromosomes. **SEE ALSO** CELL CYCLE; CHROMOSOMAL ABERRATIONS; CHROMOSOME, EUKARYOTIC; FRUIT FLY: *DROSOPHILA*; *IN SITU* HYBRIDIZATION.

antibodies immune-system proteins that bind to foreign molecules

Gail H. Vance

Bibliography

- Alberts, Bruce, et al. *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.
- Craig, J. M., and W. A. Bickmore. “Chromosome Bands—Flavours to Savour.” *Bioassays* 15, no. 5 (1993): 349–353.
- Earnshaw, W. C. “Meiotic Chromosome Structure.” *Bioassays* 9, no. 5 (1988): 47–150.
- Sumner, A. T. “The Nature and Mechanisms of Chromosome Banding.” *Cancer Genetics and Cytogenetics* 6 (1982): 59–87.
- Therman E., and M. Susman. *Human Chromosomes: Structure, Behavior and Effects*, 3rd ed. New York: Springer-Verlag, 1993.
- Yunis, J. J., and R. C. Lewandowski. “High-Resolution Cytogenetics.” *Birth Defects: Original Article Series* 19, no. 5 (1983): 11–37.

Chromosomal Theory of Inheritance

The chromosomal theory of inheritance is the idea that genes, the units of heredity, are physical in nature and are found in the chromosomes. The theory arose at the turn of the twentieth century, and became one of the cornerstones of the modern understanding of genetics.

The Birth of a Science

Charles Darwin first conceived the idea of hereditary units when he published his theory of pangenesis in 1868. In this model, circulating units called gemmules are accumulated in the gonads and transmitted to the offspring. This theory was discredited by experimental tests performed by Francis Galton in the 1870s. Galton used blood transfusions in rabbits to



show that the alleged gemmules in one rabbit's blood did not alter the heredity of the recipient rabbit's blood. In the 1890s Hugo de Vries took the term "pangenesis" and trimmed it to "pangene" for the assumed units of inheritance. He argued that pangenes remained inside the cell and did not migrate. It was this theory of intracellular pangenesis that led de Vries to independently find what Gregor Mendel had discovered thirty years earlier in his work with contrasting traits in garden peas—there are units of inheritance that are transmitted by reproduction. Wilhelm Johansson introduced the term "gene" to replace several contending and misleading terms for the basic unit of heredity in 1909. The term "genetics" came earlier, when William Bateson coined the word in 1906 to represent the new field that studied heredity, variation, and evolution. The terms "gene," "genetics," and the biblical term "genesis" all share a common Latin root, *gen*, meaning origin.

Mendel identified what he called "factors" (later called genes) as the underlying cause for the appearance of certain traits in peas. He described them as stable units that seemed to disappear in a hybrid (a plant grown from a cross between two parent plants that show differing traits) but would reappear among some of the progeny of such hybrids. Mendel identified two laws—the law of segregation and the law of independent assortment—which together governed the movement of factors from parent to **progeny**. This strongly suggested that the factors of inheritance were discrete physical objects. Shortly before Mendel's work was being rediscovered, advancements in the construction of microscopes had allowed scientists to make careful observations of cell division. This led to the discovery of colored bodies in the cell nucleus that appeared to double and divide just before each division. These were called chromosomes ("colored bodies").

By 1902 the chromosome movements during meiosis had been worked out, and Walter Sutton used them to explain Mendel's laws. He argued that the pairing and separation of **homologues** would lead to the segregation of a pair of factors they carried. Thus, to use one of Mendel's own experiments, hybridizing yellow and green pea plants would yield one yellow and one green gamete apiece. (A gamete is germ cell, sperm or egg, that contains half of a full complement of chromosomes, originating from one of the parent plants.) The result is a gametic ratio of 1:1. The union of **pollen** and **ovules** would result in the 3 yellow to 1 green Mendelian ratio. Similarly, two different pairs of homologs would yield the 9:3:3:1 ratio associated with Mendel's law of independent assortment. Sutton called his union of **cytology** with Mendelian breeding analysis the "chromosome theory of heredity."

X-Linked Inheritance in Hybrids

Beginning in 1907, Thomas Hunt Morgan extended Sutton's insights by conducting laboratory studies of the fruit fly, *Drosophila melanogaster*. With his students Alfred Henry Sturtevant, Calvin Blackman Bridges, and Hermann Joseph Muller, he established what is now called classical genetics. Morgan and his students found new phenomena that added to Sutton's chromosome theory of heredity. The first finding, achieved in 1920, was X-linked inheritance, in which white-eyed flies showed a sex-linked inheritance of the trait in a modified 3:1 ratio. In other words, cross-breeding hybrid

progeny offspring

homologues chromosomes with corresponding genes that pair and exchange segments in meiosis

pollen male plant sexual organ

ovules eggs

cytology the study of cells

red-eyed flies resulted in all the female offspring having red eyes, whereas half the male offspring had white eyes.

Morgan's team explained this modified ratio by proposing that the eye-color genes are carried on the X chromosome, of which females have two but males have only one. The female's two X chromosomes can be **homozygous** (the genes carried are AA or aa) or **heterozygous** (the genes carried are Aa) for an X-linked gene. Males are always homozygous, because the small Y chromosome lacks almost all the genes found on the X chromosome. Thus, they can carry only one gene for the trait: AY or aY. Since males can carry no second copy of the gene for the trait, they will express the white-eyed trait if they inherit the gene for it from a hybrid red-eyed parent. This discovery further strengthened the case in favor of the chromosomal theory of inheritance.

Several additional X-linked mutations arose by 1913. Morgan reported that these mutations produced unusual ratios when subjected to breeding analysis. He explained these findings by proposing that genes could trade places between two homologous chromosomes. Morgan's finding, called crossing over, was used to map the genes along the length of the X chromosome, and Sturtevant used Morgan's data to construct the first linkage map. Working independently, maize geneticists, especially Barbara McClintock, later demonstrated the same phenomenon.

Further Advances in Theory

Also from 1913 to 1916, Bridges found some exceptions to the expected modified 3:1 ratio for white-eyed flies. He inferred, and confirmed by microscopic examination of cells, that these unexpected departures arose from the failure of homologous chromosomes to separate during meiosis. Bridges called this phenomenon nondisjunction and used it as a proof of the chromosome theory of heredity.

While both Mendel and Morgan's group worked with simple, single-gene traits, the relation of genes to most character traits turned out to be more complex. A successful analysis of this was presented by Muller. He argued that the variable wing shapes and lengths of beaded and truncated wings in fruit flies involved several factors. A chief gene was essential, but it required modifier genes that could intensify or diminish its expression. By combining different modifier genes and the chief gene in two parents, Muller could predict the percentages of wing shape and length among the progeny. Muller used this analysis to support a Darwinian model of natural selection of character traits whose variations owe their origins to the highly heterozygous state of natural populations and to new mutations that arise in each generation. Muller's analysis added evolution to cytology and breeding analysis as the three tributaries of classical genetics. SEE ALSO EPISTASIS; FRUIT FLY: *DROSOPHILA*; LINKAGE AND RECOMBINATION; MCCLINTOCK, BARBARA; MENDEL, GREGOR; MORGAN, THOMAS HUNT; NATURE OF THE GENE, HISTORY.

Elof Carlson

Bibliography

Allen, G. E. *Thomas Hunt Morgan: The Man and His Science*. Princeton, NJ: Princeton University Press, 1978.

homozygous containing two identical copies of a particular gene

heterozygous characterized by possession of two different forms (alleles) of a particular gene



Judson, H. P. *The Eighth Day of Creation: The Makers of the Revolution in Biology*. New York: Simon & Schuster, 1979.

Morgan, Thomas Hunt, et al. *The Mechanism of Mendelian Heredity*. New York: Holt Reinhart & Winston, 1915. Reprinted by Johnson Reprint Corporation with an Introduction by Garland E. Allen, 1978.

Sturtevant, Alfred Henry. *A History of Genetics*. New York: Harper & Row, 1965.

Chromosome, Eukaryotic

Living organisms are divided into two broad categories based upon certain attributes of cell structure. The first category, the prokaryotes, includes bacteria and blue-green algae. Eukaryotes include most other living organisms. One of the most important features distinguishing eukaryotes from prokaryotes is the chromosomal arrangement of genetic information in the cells. Eukaryotes enclose their genetic material in a specialized compartment called the nucleus. Prokaryotes lack nuclei.

Basic Organization

In 1883, Wilhelm Roux proposed that the filaments observed when cell nuclei were stained with basic dyes were the bearers of the hereditary factors. Heinrich Wilhelm Waldeyer later coined the word chromosome (“colored body”) for these filaments. The eukaryotic chromosome now is defined as a discrete unit of the genome, visible only during cell division, that contains genes arranged in a linear sequence. Eukaryotic organisms contain much more genetic information than prokaryotes. For example, the eukaryotic organism *Saccharomyces cerevisiae* (baker’s yeast) contains 3.5 times more DNA in its **haploid** state than the prokaryotic *Escherichia coli*, while higher vertebrate cells contain more than 1,000 times the DNA.

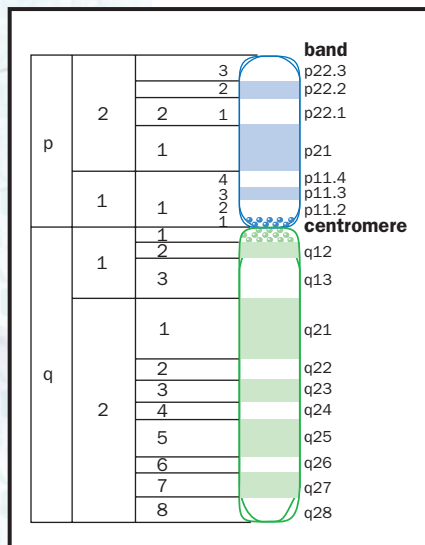
The basic component of the eukaryotic chromosome is its DNA, which contains all of the genetic material responsible for encoding a particular organism. Genes are arranged in a linear array on the chromosome. A major distinction between eukaryotic and prokaryotic chromosomes is that eukaryotic chromosomes contain vast amounts of DNA between the genes. The function of most of this “extra” DNA is unknown. It contains repetitive sequences, functionless gene copies called pseudogenes, transposable elements, and other types of DNA.

Eukaryotic genes may be dispersed randomly throughout the chromosome or they may be specifically organized. A gene family is a set of genes that originated from the duplication and subsequent variation of a common, ancestral gene. Members of a gene family may be clustered on the same chromosome, as in the case of the globin genes. Gene duplication events also have resulted in gene clusters in which related or identical genes are arranged in tandem. Examples of gene clusters include the genes for rRNA and histone proteins.

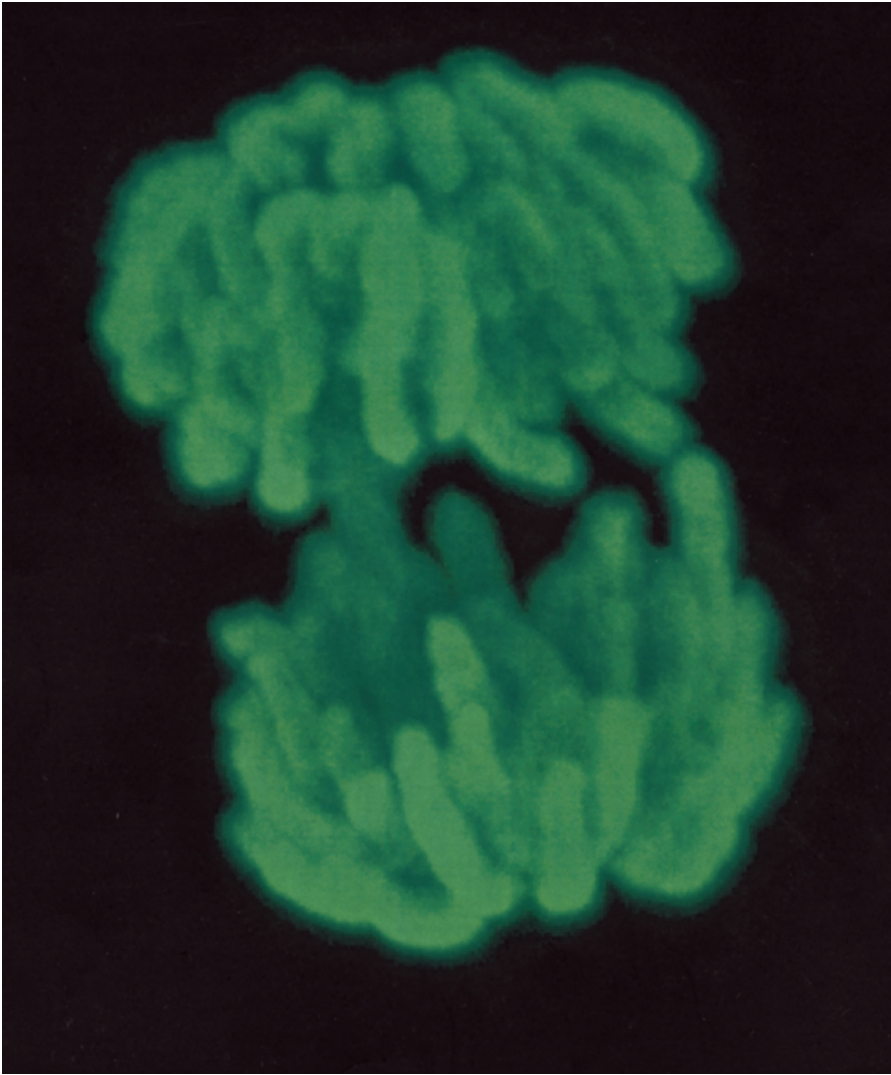
Higher-Order Organization

The DNA of a eukaryotic cell must be constrained within the confines of the nucleus. In human cells, six billion base pairs are contained on the forty-six chromosomes of double-stranded DNA. This DNA has a total length of

haploid possessing only one copy of each chromosome



Chromosome staining reveals distinct bands that can be numbered for reference. Adapted from Lewin, 1997.



Eukaryotic chromosomes separating during the later stages of cell division.

1.8 meters, yet it must fit into a nucleus with an average diameter of 6 **micrometers**. This feat is accomplished in part by the packaging of DNA into chromatin, a condensed complex of DNA, histones, and nonhistone proteins.

The basic unit of chromatin is the nucleosome. The nucleosome is composed of approximately 146 base pairs of DNA wrapped in 1.8 helical turns around an eight-unit structure called a histone protein octamer. This histone octamer consists of two copies each of the histones H2A, H2B, H3, and H4. Nucleosomes form arrays along the DNA. The space in between individual nucleosomes is referred to as linker DNA, and can range in length from 8 to 114 base pairs, with 55 base pairs being the average. Linker DNA interacts with the linker histone, called H1, and there are equal numbers of nucleosomes and H1 histone molecules in the chromatin. A nucleosome particle bound to a single molecule of H1 is termed a chromatosome.

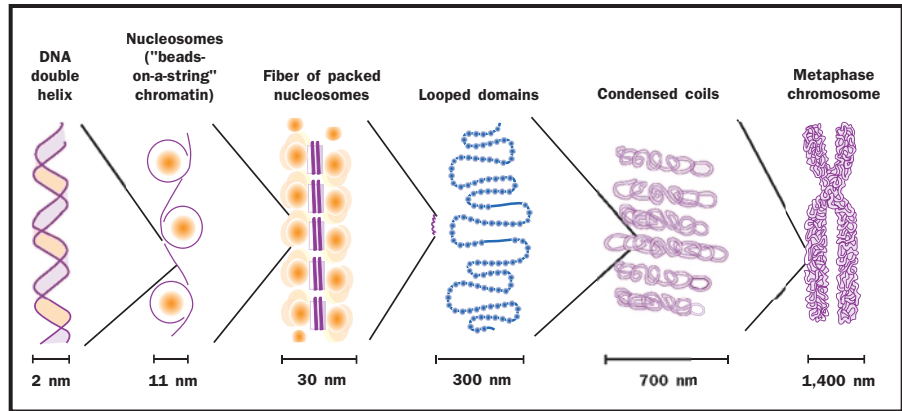
Higher-order chromatin structure can be visualized microscopically as fibers 10 and 30 **nanometers** in diameter. The 10-nanometer fiber can be observed under conditions of low ionic strength. This fiber resembles beads

micrometer 1/1000 meter

nanometer 10^9 meters; one billionth of a meter



Levels of organization of DNA in the chromosome. The double helix is wound around histone proteins to form nucleosomes. Condensed coils characterize regions that are not expressed. The metaphase chromosomes only appear during cell division. Adapted from Curtis, 1989.



in vivo "in life"; in a living organism, rather than in a laboratory apparatus

in vitro "in glass"; in lab apparatus, rather than within a living organism

on a string, and is in fact a string of nucleosomes. The structure does not require the presence of histone H1. It is unclear if the 10-nanometer fiber exists **in vivo** or if it is just an artifact of chromatin unfolding during extraction **in vitro**.

Cellular chromatin usually exists as a 30-nanometer fiber. It can be seen under conditions of higher ionic strength. The presence of the linker histone H1 is required for the formation of this fiber, as it helps promote compaction and condensation. The 30-nanometer fiber is formed into a coil, but its exact structure has not been determined.

During cell division, chromosomes condense to an even greater extent. The mechanism by which the 30-nanometer fibers are packed into the highly condensed, organized structure of the mitotic chromosome is unclear. The compaction of chromosomes during cell division is accomplished in part by the organization of chromatin into large, looped structures that are attached at their bases to a protein scaffold. This scaffold remains intact even if the DNA is experimentally removed. It is possible that this scaffold is responsible for maintaining the shape of the chromosomes.

Heterochromatin versus Euchromatin

Chromatin can be divided into two regions, euchromatin and heterochromatin, based on its state of condensation, that is, based on how tightly its constituent elements are packed together. Most of the cellular chromatin is euchromatin, which has a relatively dispersed appearance in the nucleus. It condenses significantly only during mitosis. Genes within euchromatin can be transcriptionally active or repressed at a given point in time.

Heterochromatin, on the other hand, is condensed in **interphase**, usually does not contain genes that are being expressed, and is among the last portions of the genome to be replicated prior to cell division. Heterochromatin frequently is localized at the periphery of the nucleus. It can be subdivided into constitutive and facultative heterochromatin. Constitutive heterochromatin is always inactive. It is often found adjacent to centromeres and telomeres. Facultative heterochromatin refers to DNA sequences that are specifically inactivated as the result of development or a regulatory event. One example of facultative heterochromatin is the mammalian X chromosome. The single X chromosome present in male cells is active. However,

interphase the time period between cell divisions

in female cells, one of the two copies present is directly and specifically inactivated.

Cytological Features

While the interphase chromatin appears to be a tangled mass within the nucleus, the mitotic chromosome appears as an organized structure with many prominent features. These features include structures known as the **centromere** and the **telomere**.

The centromere is the region of the chromosome to which the spindle apparatus attaches during mitosis and meiosis. The spindle apparatus is the network of fibers along which the chromosomes move during cell division. It also contains the site at which sister chromatids are attached prior to their separation during the stage of cell division known as anaphase. The centromere is responsible for the movement of the chromosome. During mitosis and meiosis, the centromere is pulled by the spindle fibers toward the opposite ends of the dividing cell (poles), as the attached chromosome is dragged behind. The centromere is essential for segregation.

The telomere is a structure that occurs at the end of linear eukaryotic chromosomes and that confers stability. The first telomere to be sequenced was from the organism *Tetrahymena thermophila*, a type of single-cell eukaryotic organism, in 1978 by Elizabeth Blackburn and Joseph Gall. This telomere contains an AACCC nucleotide sequence that is repeated thirty to seventy times. The sequences of telomeres from other species show the same pattern: a tandem array of a short nucleotide sequence, one DNA strand G-rich and the other DNA strand C-rich. Telomeres are synthesized by an enzyme called telomerase, which adds telomeric sequences back onto chromosome ends, one base at a time.

Banding techniques allow every mitotic chromosome, as well as regions within individual chromosomes, to be distinguished. The first method used, known as Q-banding, uses fluorescent derivatives of quinacrine, which for unknown reasons bind preferentially to some regions of chromosomes. When viewed under ultraviolet light, chromosomes appear with bright bands, corresponding to euchromatin, and dark bands, corresponding to heterochromatin. This banding method is useful in identifying some polymorphisms, which are gene variations (**alleles**) within the population of genomes. It is also useful for identifying the Y chromosome.

Later, another banding technique was developed, called G-banding. This technique employs a modified Giemsa stain, which is a dye that specifically binds to DNA. As in Q-banding, a series of identifiable light and dark bands is generated on the chromosome. Euchromatin stains lightly, while most heterochromatin stains darkly.

These banding methods have made it possible to diagrammatically represent each human chromosome, using designated nomenclature for specific chromosomal regions. These techniques have shown that the mitotic chromosome can be divided into a short arm, designated p, and a long arm, q. Each arm is then divided into one to three regions by landmarks, such as the ends of the arm, the centromere, and certain prominent bands. Regions are spaces between adjacent landmarks and are numbered consecutively within each region. These features allow genes to be designated to specific

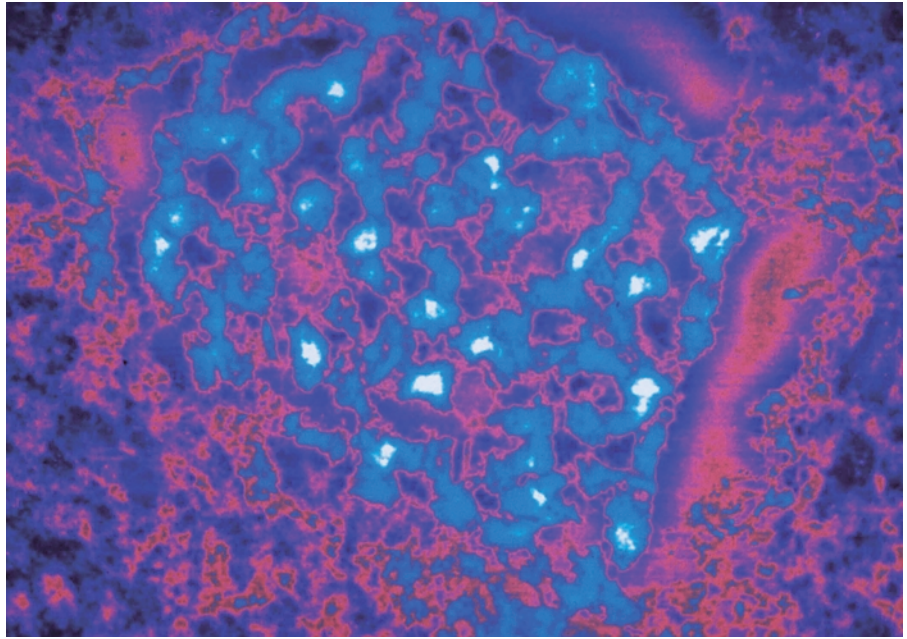
centromere the region of the chromosome linking chromatids

telomere chromosome tip

alleles particular forms of genes



Giant chromosomes from a fruit fly larva's salivary glands enable study of RNA transcription. The faint, "puffed" areas are the sites of active transcription.



regions on the mitotic chromosome. For example, a gene at band 4q14 is localized to chromosome 4, the long arm, region 1, band 4.

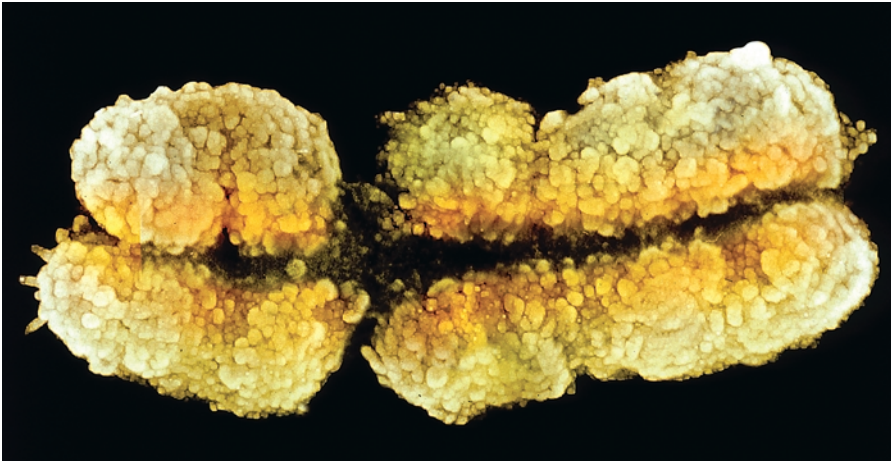
Polytene Chromosomes

Due to the compact nature of chromatin in eukaryotic cells, it is virtually impossible to visualize gene expression *in vivo*. However, there are certain unusual situations in which gene expression can be seen. Such is the case with the polytene chromosomes, which are exceptionally large in comparison to other types of chromosomes.

The salivary glands of *Drosophila melanogaster* (fruit fly) larvae contain greatly enlarged chromosomes. These are polytene chromosomes, and they result from multiple rounds of replication of a diploid pair of chromosomes joined in parallel. The replicated chromosomes remain attached to one another. Each pair of chromosomes can replicate up to nine times; thus, the resultant polytene chromosome can contain up to 1,024 (2^9) strands of DNA.

The vast majority (95%) of DNA in the polytene chromosomes is concentrated in chromosomal bands, called chromomeres, which are microscopically visualized through staining. These chromomeres form a pattern that is characteristic for each *Drosophila* strain. *Drosophila* polytene chromosomes display roughly 5,000 bands. Since the total number of genes in *Drosophila* appears to be greater than the number of bands that can be visualized, it is likely that there are multiple genes located within a given band.

The banding pattern of the polytene chromosomes provides a cytological map, or diagrammatic representation, of the physical location of genes at specific sites in the cell. The positions of individual genes can be determined using a technique called *in situ* hybridization. First, the DNA of an immobilized chromosome preparation is made single-stranded (denatured). A radioactively labeled probe, generally a small piece of DNA corresponding to the gene of interest, is then mixed with the denatured DNA under conditions that permit the radiolabeled DNA to bind to its complementary



Scanning electron microscope magnification of one human X chromosome. Genetic information is packed within the “bumps” and loops of material pictured here.

DNA strand on the immobilized chromosome preparation. Finally, the chromosomal binding site is determined by way of a procedure called autoradiography, which is used to make the radioactive probe visible on photographic film.

With autoradiography, the sites of gene expression can be visualized along the polytene chromosomes. As DNA decondenses into a more open state, it forms a distinctive swelling known as a chromosomal puff. These puffs are active sites of transcription, or RNA synthesis. Throughout development, they alternately expand and contract, as specific genes are activated or repressed.

Chromosome Organization, Replication, and Transcription

The compact nature of chromatin structure presents a barrier to processes that require access to DNA, such as replication and transcription. It seems likely that the separation of parental DNA strands during replication must disrupt higher-order chromatin structure to at least some degree. In the *Drosophila* polytene chromosomes, this disruption can be seen, at least in part, as puffs at sites of DNA replication. It is unclear, however, what happens to the nucleosomes during this process. If the nucleosomes are removed during replication, they are quickly reassembled, for there is no time period during which microscopic analysis shows the DNA to be free of nucleosomes.

Transcriptional activation of eukaryotic genes requires that the machinery responsible for synthesizing the RNA gain access to the regulatory regions of the DNA that control gene expression. This again necessitates some decondensation of the chromatin structure. Decondensation can be facilitated by protein complexes known as chromatin-remodeling **enzymes**. Chromatin-remodeling enzymes alter the structure of chromatin in such a way that regulatory factors can gain access to the DNA. These enzymes are divided into two groups, those that chemically modify histones and those that use the energy derived from ATP hydrolysis to alter histone-DNA linkages.

enzymes proteins that control a reaction in a cell

Chemical Modification of Chromatin Structure

The best-characterized of the enzymes capable of chemically modifying histones to open chromatin structure are the histone acetyltransferases, or



HATs. These enzymes add acetyl groups to lysine residues within the amino termini (also known as the tails) of H3 and H4 histones. This adds a negative charge to the histone tails. The negative charge is believed to cause them to push away from the DNA backbone, resulting in a somewhat less condensed chromatin structure. Hyperacetylation of histones in the promoter regions of genes is associated with active, ongoing gene expression, while histone hypoacetylation is associated with genes that are transcriptionally silent.

The association between histone acetyltransferases and gene activation was first suggested when it was found that the *Tetrahymena thermophila* HAT p55 was structurally similar to the yeast protein GCN5. GCN5 previously had been shown to be involved in gene activation. Later it was discovered that GCN5-regulated genes are hyperacetylated when active, and that certain mutations in GCN5 that affect its ability to activate target genes result in diminished levels of acetylation of these regions. Histone acetyltransferases also are found in mammalian cells. It is believed that the HATs are recruited to specific genes by specific transcriptional activators.

The activity of the histone acetyltransferases is opposed by histone deacetylases. These enzymes remove acetyl groups from the histone tails, resulting in the repression of gene activation. As with the HATs, these enzymes are believed to be recruited to chromatin by repressor proteins to aid in the inactivation of specific genes.

Transcriptional activation also can be repressed by the methylation of specific cytosine residues found in some genes. It is unclear how methylation results in transcriptional repression, but it is known to cause the inhibition of specific transcriptional activators and to initiate the recruitment of specific repressors that bind methylated DNA. At least one methylated DNA-binding repressor can be isolated from cell extracts together with (and therefore appears to be associated with) histone deacetylases, thereby providing a link between methylation and deacetylation and gene inactivation. Other enzymes chemically modify histones by adding or removing phosphate, ubiquitin, and other chemical groups.

ATP-Dependent Chromatin-Remodeling Complexes

The presence of enzymes that can alter the structure of chromatin was suggested by yeast genetic studies that identified a number of genes, called SWI and SNF genes. These genes are required for multiple transcriptional activation events. A key breakthrough came when it was discovered that yeast cells could compensate for a deficiency in these SWI and SNF gene products by altering their chromatin structure. This led to the hypothesis that SWI and SNF genes are involved in the regulation of chromatin structure. It is now known that SWI and SNF proteins form a large, multisubunit complex, termed SWI/SNF, that can hydrolyze ATP and use the energy thus generated to alter chromatin structure.

Similar proteins that can hydrolyze ATP are present throughout the eukaryotic kingdom, and these form related multiprotein enzymes that also possess chromatin-remodeling properties. The mechanism by which these complexes alter the chromatin structure is unclear, but it is likely that the enzymes break or loosen the linkage between histone and DNA in a manner that increases the mobility and flexibility of the DNA wrapped around